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Price promotions and "freemium" app monetization

Julian Runge[1,2] · Jonathan Levav[3] · Harikesh S. Nair[3]

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Abstract

The "freemium" model for digital goods involves selling a base version of the product for free, and making premium product features available to users only on payment. The success of the model is predicated on the ability to profitably convert free users to paying ones. Price promotions (or "sales") are often used in freemium to induce this conversion. However, the causal effect of exposing consumers to such inter-temporal price variation is unclear. While sales can generate beneficial short-run conversion, they may be harmful in the long-run if consumers inter-temporally substitute purchases to periods with low prices, or use them as signals of low product quality. These long-run concerns may be accentuated in freemium apps, where the base version is sold for free, so that sales form extreme price cuts on the overall product combination. We work with the seller of a free-to-play video game to randomize entering cohorts of users into treatment and control conditions in which promotions for in-game purchases are turned on or off. We observe complete user behavior for half a year, including purchases and consumption of in-game goods, which, in contrast to much of the extant literature, enables us to assess possible substitution over time in consumption directly. We find that conversion and revenue improve in the treatment group; and detect *no evidence* of harmful inter-temporal substitution or negative inferences about quality from exposure to price variation, suggesting that promotions are profitable. We conjecture that the zero price of the base product that makes its consumption virtually costless, combined with the complementarity between the base product and premium features can help explain this. To the extent that this holds across freemium contexts, the positive effects of promotions documented here may hold more generally.

We thank the video game company on whose mobile video game app we implemented the experiment reported in this paper. The company wishes to remain anonymous. No confidential information, user details or trade secrets are revealed in this paper. All opinions in this paper represent the views of the authors and not of the company or the respective universities.

✉ Julian Runge
julian.runge@duke.edu

Extended author information available on the last page of the article.

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JEL Classification M31 · L11

1 Introduction

This paper focuses on the role of promotions in facilitating a "freemium" business model for mobile apps. In a freemium model, firms offer a version of their app for free to acquire and engage with consumers, and then "upsell" premium upgrades that require payment. In mobile gaming apps, for instance – the focus of the empirical application in this paper – the base version of the game is typically free-to-play, and users can pay to purchase in-game goods, which can be used to unlock a desired level of upgrades or to proceed to more advanced levels of the game. Based on recent estimates, 54% of all apps on the Google Play and the Apple store in 2019 utilize this business model (Fyber, 2020). The success of the freemium model relies on the efficiency with which free users are converted to paying ones and, for this reason, promotions can be a key enabler of the model. Promotions help conversion by bridging the so-called "penny gap" in freemium – a colloquial term to describe a commonly held belief in the start-up community that it is harder to convince a customer to pay the first penny than to induce him to pay more once initial payment has been made (Anderson & Simester, 2010; Shmilovici, 2011; Carter, 2019).[1] In gaming, apart from improving free-to-pay conversion, promotions can also increase repeat purchases of in-game goods, facilitating unlocking of premium features that improve user experience and increase future retention and gameplay. As such, they are common in freemium apps, and the conventional wisdom is that offering promotions is beneficial for the seller; a decrease in price increases demand.

The motivation for this study is the considerable skepticism with which extant literature would suggest to view this conventional wisdom. While short-run conversion likely increases during promotions, whether the long-term consequences of periodic promotions or "sales" are beneficial to the seller is far from obvious. The concern is that in situations with repeat purchase and in which consumers have price knowledge, systemic price variation can cause consumer behavior to adjust so as to "game the system." For instance, consumers who anticipate the promotion cycle can time their purchases, delaying current purchases and pulling forward future purchases to low-price sales periods. Further, as most digital goods are storable at low costs, users could easily stockpile promoted goods at low prices and consume out of this inventory in future high price periods. By this view, sales can become expensive giveaways, essentially serving to generate unprofitable intertemporal demand substitution,

[1]The difficulty of converting free users to paying ones could arise from a fixed cost to the user of setting up payment (e.g., the customer has to enter his credit card details and get verified) or by the special significance of a "zero price" in consumer's minds (Heyman & Ariely, 2004; Shampanier et al., 2007; Ascarza et al., 2012).

moving a purchase that would otherwise have occurred at a high-price future or past to a low-price present featuring a sale. Indeed, for this reason a body of work in the academic literature has warned that the conventional wisdom about the advantage of promotions must be evaluated with caution (Mela et al., 1997; Nijs et al., 2001; Erdem et al., 2003; Hendel & Nevo, 2003; Anderson & Simester, 2004; Neslin & van Heerde, 2009; Anderson & Simester, 2010; Nair et al., 2017; Elberg et al., 2019).

Further, in situations where products are "experience goods" and consumers are uncertain about their match-value with the product, a separate literature has warned that consumers can use low prices as a signal of correspondingly low quality (Gerstner, 1985; Milgrom & Roberts, 1986; Rao & Monroe, 1989; Erdem et al., 2008; Dubé et al., 2017). If this occurs, exposure to sales can reduce user satisfaction and product usage, harming the seller. The concern about sales signaling quality in freemium arises if users evaluate the "base good + premium features" as a system of goods they are consuming, and then view the price cuts on the premium features during the promotion as price reductions on this full system. Since the base version is already being provided for free, additional price cuts on premium features may be viewed as extreme price cuts on the full system, leading to concerns about negative quality perceptions. Additionally, in several app contexts, some of the premium features may be attainable via continued utilization of the base product itself without payment. For instance, in mobile gaming, many premium features that users pay to acquire could alternatively be obtained for free, albeit very slowly, by spending time and effort in the base game or by watching ads. Then, it is possible users view premium features as components of the base product itself, so that price cuts on the premium features during promotions affect their quality perceptions of the base product.

While these reasons make us skeptical about accepting the conventional wisdom at face value, there are also countervailing reasons why the effects emphasized in this literature may play out differently in freemium app environments. The research that casts doubt on the long-term value of sales mostly draws its conclusions from studies of non-digital, storable consumer packaged goods (CPG – e.g., paper towels, potato chips, coffee, etc.) and often in offline store environments. The purchase of such goods is typically separated from their consumption – a consumer that purchases three bags of chips on sale is highly unlikely to consume them all during the store visit – so that stocking up and purchase planning are a viable alternative response to a sale. In addition, the goods that have been considered in this literature have typically not been ones that feature complementarities or ones whose consumption is enhanced by joint consumption with other goods. In contrast, the freemium app environment and the digital goods sold in them differ from typical durable or consumer packaged goods in several respects. The freemium model features two types of goods – the free, "base" good, and the paid, premium features or "add-on" goods.[2] Upgrades to added-on premium features are ordinarily executed in the course of the base good's consumption, as a result of an immediate need to enhance the base product or due

[2]In the context of gaming apps, we can think of the base version of the game as the base good and the premium features as the add-on goods. Spending in-game currency allows the user to acquire add-on goods.

to an impulse, which may make immediate consumption of the purchased add-on more likely, reducing stockpiling. Further, the base product and premium features tend to be *complements* to each other. Higher consumption of the base product makes it more likely that premium features are added-on; and higher consumption of premium features likely makes the base product more attractive. Taken together, these aspects may make saving the currently purchased premium features for the future less likely or unattractive, especially for high-value users who respond to promotions by buying more of the add-on goods. Consequently, the interplay of price variation and stockpiling behavior that occurs in freemium app environments is an empirical question.

Motivated by these considerations, we implement a field experiment that formally investigates the impact to a freemium app seller of introducing its users to periodic promotions. The randomization induced by the experiment enables us to causally assess the net, long-run effect effect of promotions in such a setting. Using the experiment, we also analyze the extent to which intertemporal substitution and/or negative quality perceptions undermine the efficacy of promotions. Finally, leveraging access to user-level data, we explore how user consumption of base and add-on goods responds to promotions, so as to provide a better understanding of how promotions work in such settings.

Our field experiment is implemented in collaboration with a video-game company that is active in the mobile gaming sector. The mobile gaming sector is particularly suited for our purposes as it is the largest source of mobile app monetization: In 2019, gaming apps accounted for 69% of $84B global app store consumer spending (Sensortower, 2020). Honing in on the two largest app stores – `Google Playstore` and `iOS Appstore`, gaming accounted for about 83% and 66% of in-app revenue respectively (see again Sensortower, 2020). The mobile gaming sector is also one that is typically monetized using a freemium model and one that features heavy use of promotions.[3] The mobile gaming industry is a sophisticated adopter of experimentation, facilitating our ability to implement a randomized control trial within our focal app, which enables a credible evaluation of the long-run impact of exposure to periodic promotions. Finally, the environment of a video game allows us to track large amounts of granular, anonymized user data on base and premium product adoption, spending and consumption and to analyze user behavior within a controlled environment. This helps improve statistical precision, for picking up a set of effects that are less likely to be affected by factors outside of the experiment, and for exploring the underlying mechanisms for the effects we observe.

The game chosen for the field experiment combines a puzzle with a city building component and is representative of the free-to-play game genre. It is published on `Google`'s, `Apple`'s and `Facebook`'s app marketplaces and has been downloaded by more than 50 million players as of October 2017, when the experiment ended. The base version of the game is free-to-play to all users. Users can buy

[3] In-game purchases in free-to-play games accounted for 82% of worldwide digital games revenue in 2017 (Gough, 2018). The market for mobile in-game consumer spending alone makes up close to 80% of $101 billion in-app revenue in 2018 (App Annie, 2018).

in-game currency using real money, which they can use to unlock a variety of premium game features. The question of pricing pertains to the exchange rate between in-game currency and real money. Price variation is induced when the exchange rate for in-game currency is discounted via periodic promotions. Starting mid-December 2016, we randomize cohorts of consumers who download the game app into treatment and control conditions in which their exposure to price variation is switched on or off. The treated group is exposed to a fixed schedule of periodic promotions. The schedule features "Hi-Lo" pricing analogous to sales in CPG settings (Hoch et al., 1994; Bell & Lattin, 1998; Ho et al., 1998; Ellickson & Misra, 2008; Ellickson et al., 2012 – specific details are described later in the paper). The control group sees no promotions. Individual-level data on game usage and spending for the users are tracked for about half a year (180 days) following app download.

Analyzing the data, we find that exposure to the promotion sequence generates large benefits to the firm. Relative to the control group, conversion increases by 37.1%; purchases of in-game currencies increase by 27.2%; and revenue from premium features increases by 23.6%. Surprisingly, we detect no evidence of harmful intertemporal substitution of demand for in-game currency. Average outcomes pre- and post-promotion cycles are statistically indistinguishable between the treated and control groups. Thus, it appears that almost all the lift observed during promotion cycles represents incremental expansion of demand and increased consumption of purchased goods. We also see little evidence of prices adversely signaling quality. Usage of the game amongst users remains statistically unchanged between the treated and control group. This result indicates that promotions do not serve as a strong quality signal that would lead users to abandon the app.

We then explore plausible explanations for the observed lack of intertemporal substitution. We postulate that this result can be explained by complementarities between the base and added-on goods, along with positive correlation in user valuations for the base and add-on goods as it provides a reason for users who buy add-on goods during promotions to consume them almost immediately with little stockpiling. The rationale is explained in more detail in the paper, but in short is as follows. As users consume the add-on goods bought in a promotion, their utility derived from time spent using the base product jumps up, enticing them to further consume the add-on goods, leading to further consumption of the base good, and so on, leading to high levels of consumption of both in the immediate vicinity of a promotion. Furthermore, this effect is higher when those that value the base good more also value the add-on goods more. The empirical patterns we uncover in users' consumption of base and add-on goods in response to promotions are consistent with this explanation.

To the extent that the immediacy of utility increases from deal purchases and this pattern of heterogeneity and complementarity between the base and premium products apply in other freemium environments, price variation in the form of sales can be a profitable policy more widely. Complementarity and positive correlation in user valuations for the base and add-on goods plausibly exists in many freemium contexts. For instance, in `Dropbox`, additional collaboration opportunities (a premium feature) may make the storage space available in the base version more valuable, and vice versa, albeit in a more paced manner. This is a form of complementarity. Further, those who value collaboration (e.g., business users) may also value the base version

more. This implies positive correlation in valuations. While we think these patterns are plausible in many contexts, we caution the reader that we do not mean to imply that the specific quantitative results uncovered in this experiment generalize across all scenarios found in the digital app economy. Further research and inquiry on these aspects is warranted to generalize beyond our context. Our results and explanation based on this experiment can serve to represent an approach and a set of hypotheses on the basis of which to proceed on this line of inquiry.

The rest of this section closes with a more detailed review of the related literature. Then, we describe the empirical setting and field experiment. Subsequently, we discuss the results from the experiment. The last section concludes.

Relationship to the literature On the data side, the digital app-based environment facilitates some novel aspects of this study in studying sales: (a) inducing randomization and controlling experiences at the user-level, which improves statistical power and facilitates exploration of heterogeneity in user-level response; (b) sustaining a control group with no promotions for a long period of time (180 days), which has typically been difficult in many settings due to the costs of such experimentation and the inability to closely control user experiences; and (c) observation of consumption and purchases over time at the user-level, which directly facilitates the study of users' substitution of goods over time. While past papers have leveraged these aspects in isolation, to our knowledge none have brought all three aspects together in a study of intertemporal pricing. The present paper is also the first that has presented an evaluation of sales in a digital, freemium setting using a randomized controlled trial. Further, we observe consumption, enabling a direct assessment of intertemporal demand substitution. This direct assessment is novel compared to past literature studying promotions and intertemporal substitution in storable goods, which has typically observed purchases but not consumption, and has therefore relied on indirect assessments.

The paper is related to a stream of papers that leverage price randomization to study static price discrimination, e.g., Levitt et al. (2016), Sahni et al. (2016), Dubé et al. (2017), and Dube and Misra (2017). By and large, price randomization is much rarer than, say, advertising randomization, with the associated stream of literature comprising only a handful of studies. Our study is distinct from the papers in this stream as it focuses on measuring the causal effect of sustained exposure to a *sequence* of prices over time to investigate its intertemporal consequences for initial and repeat purchases. Levitt et al. (2016)'s study which involves virtual goods in a video game is closest to our work. Their interest is in studying the effect of quantity discounts rather than intertemporal pricing, which is the focus here. We further elaborate the differences between their findings and our results later in the paper.

This paper is also related to experimental studies by Hoch et al. (1994) and Elberg et al. (2019) who implement store- or category-level randomization of pricing policies to study promotions in grocery retail (comparing "Every Day Low Pricing" versus "Hi-Lo" promotions in Hoch et al. (1994) and "Hi-Lo" promotions with deep versus shallow discounts in Elberg et al. (2019)). Given the grocery store setting, these studies do not implement a control group with "no promotions," and are not specifically focused on the interaction of promotions with a tiered system of goods.

In contrast, this study relates to digital, freemium goods that comprise base and premium tiers; involves a control group with no promotions (and hence is able to benchmark against a no-promotion environment); and implements randomization at the individual-level (which yields individual-level data and more statistical power).

Also closely related are field experiments by Anderson and Simester (2004) and Anderson and Simester (2010), which compare catalog purchases between individuals randomized into receiving a catalog with deep or shallow discounts. Anderson and Simester (2004) and Anderson and Simester (2010) find positive long-run impact from exposure to deep discounts for new consumers, consistent with our findings for new users in the studied environment. Our study is distinguished from their work by its focus on digital goods; by having access to a control condition with no discounts; and by its focus on studying the treatment effect of a pricing policy that comprise a sequence of varying prices over time, rather than a one-time reduction.

2 Empirical setting and field experiment

2.1 Empirical setting

The video game on which the experiment is implemented can be described as involving "solving puzzles and building a city."[4] Within the game, the user confronts graphical puzzles that have to be solved while trying to build a city. The puzzles and the city the user is building are connected to an overarching storyline and metagame. This induces programmatic buildup and continued engagement as gameplay progresses. Free play is possible in perpetuity and the game can be completed without a single purchase. However, if the user chooses to, she can purchase a variety of in-game premium goods to enhance gameplay. Examples of enhancing gameplay include extending a game session or using decorations to embellish the city the player is building. To purchase in-game goods, the user needs in-game currencies. In-game currencies can either be earned by gameplay, or be bought from a game shop within the app by exchanging real money.

There are three types of in-game currencies: energy, coins and cash. Energy is required to continue gameplay. Coins are used to enhance gameplay. Cash can be used to enhance gameplay, advance to higher levels, and to acquire coins; therefore, cash is more valuable than energy or coins. All users start with an initial endowment of all three currencies. Coins and energy are earned through gameplay and are replenished often (with the difference that energy is earned from "typical" gameplay, while coins are earned from "advanced" gameplay). However, cash is typically not earned nor replenished by gameplay and has to be purchased using real money.[5] Purchases of all three in-game currencies can be made from the game shop at any

[4] See https://en.wikipedia.org/wiki/Puzzle_video_game for descriptions of this genre.

[5] A small amount of cash can be earned in rare situations within the game, such as when solving a specially designed puzzle. However, such situations are highly infrequent and the cash earned this way is minuscule. Such currency configurations are typical for the app game genre (e.g., see https://en.wikipedia.org/wiki/Virtual_economy#Types_of_virtual_currencies, accessed Dec 22, 2020).

time. Purchases can be done separately or in bundles. The in-game currencies do not expire, and are exchangeable for in-game goods at rates determined by the game seller. In-game currencies are promoted to users by offering short-lived discounts on the regular price.

In addition, users are also sometimes shown in-game ads, typically video-ads by other third-party apps. Users who watch the video-ads are rewarded a small amount of in-game currency. On seeing the ad, they are shown a still-screen with a link to the app store.

Advertising makes up only a small percentage of the revenue from the product ($< 10\%$); the bulk of the revenue is from purchases of in-game currency. This monetization model is typical in the free-to-play app gaming industry (see Tomic (2019) for an overview and Fig. 10 in the Appendix for an example).[6]

2.2 Experimental design

To implement the experiment, we isolate a sub-population of new users who download the game application from Google's Playstore to their Android smartphone or tablet between December 15, 2016 and February 21, 2017. Upon download, users are randomized to a treatment group or a control group as follows.

Treatment group

- Each user in the treatment group is exposed to a *sequence* of promotions for about half a year (180 days) from the time she downloads the game.
- The promotions pertain to a *bundle* of in-game currencies offered to the user within the game.
- The promotion plan is as follows: A promotional period begins every second Friday and lasts 3−5 days. We refer to these as promotional cycles. During a promotional cycle, the user is shown a single *discounted bundle* comprising the three in-game currencies. The discount on the bundle relative to buying the same in-game currencies from the game shop is substantial, ranging from 50% to 85%. The bundles are chosen from a set of pre-fixed configurations shown in Table 1. Looking for instance at the first row of Table 1, the bundle offered at \$1.99, comprising 10 units of energy, 1,000 coins and 100 units of cash represents an 85% discount relative to the lowest price that would have to be paid to acquire the same units of currencies from the game shop. In non-promotional periods, all in-game currencies are sold at fixed base prices in the game shop.
- The specific bundle a user is shown in a promotional cycle is picked from amongst the configurations in Table 1 on the basis of the recency and monetary value of her past purchases. The behavioral rule followed by the firm's pricing team for this purpose is shown in Table 2. Broadly speaking, cheaper bundles are shown to users with smaller and less recent past purchases and higher percentage

[6]It is also representative of the "app economy" where advertising only drives a small share of revenue (Ghose and Han, 2014).

Table 1 Discounted In-Game Currency Bundles Offered During Promotions

	Bundle configuration			
Price charged for bundle	Energy	Coins	Cash	Perc. discount
$1.99	10	1000	100	85%
$2.99	10	1000	100	80%
$4.99	10	1500	100	70%
$9.99	10	2000	150	55%
$14.99	20	2000	200	55%
$29.99	50	2500	500	55%
$59.99	100	5000	1000	55%
$99.99	100	8000	1500	50%

Each row in the table shows a possible configuration of the discounted bundles offered on promotion to users in the treatment group in the experiment. The percentage discount is relative to the cheapest price that would be incurred to buy the same units of currencies included in the bundle from the game shop

discounts are offered to non-buyers than buyers. A user can only make one promotional purchase per cycle, and the length of the promotional cycle is common across all users. Apart from offering the bundle, no other price reductions of the in-game currencies are offered.

- When the promotion is active, it is advertised to the user with an in-app "pop-up." The user can choose to purchase the bundle offer by clicking on the pop-up or send it to the background. If sent to the background, it will remain as an icon on the game app's main screen until purchase or the end of the promotional cycle. Once a purchase occurs or the cycle ends, the icon disappears from the user's screen. Figure 1 presents an illustration of the sequence using the 14-day period from January 6, 2017 to January 20, 2017 as an example.

Table 2 Bundle Offered During Promotional Cycle as a Function of User's Recency and Monetary Value

Days since last purchase	Highest per-transaction amount in a past purchase							
	No purchase	(0,$1)	[$1,$4)	[$4,$8)	[$8,$13)	[$13,$26)	[$26,$50)	[$50,)
No purchase	$1.99	–	–	–	–	–	–	–
[0-8)	–	$2.99	$4.99	$9.99	$14.99	$29.99	$59.99	$99.99
[8-17)	–	$2.99	$4.99	$9.99	$14.99	$29.99	$59.99	$99.99
[17-61)	–	$2.99	$4.99	$4.99	$9.99	$14.99	$29.99	$59.99
[61-90)	–	$1.99	$2.99	$4.99	$4.99	$9.99	$14.99	$14.99
≥90	–	$1.99	$2.99	$2.99	$2.99	$4.99	$4.99	$4.99

Each row in the table shows the bundle that is offered on promotion to a user in the treatment group in the experiment as a function of the recency and monetary value of the user's past purchases on the app

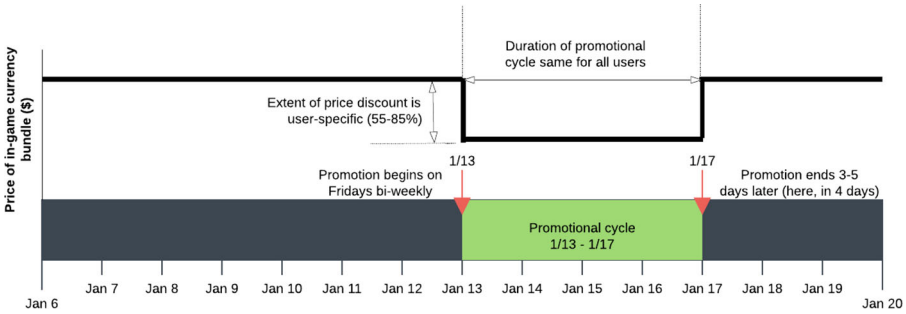


Fig. 1 Promotion Schedule in Treatment Group: Example for 14-day period from January 6, 2017 to January 20, 2017. The promotion sequence that treated users are randomized into involves "Hi-Lo" pricing: In non-promotional periods, the in-game currency is sold at a fixed base price. Every second Friday, a discounted bundle is shown to the user. The specific bundle picked depends on the user's past buying history. The discount offer is also personalized to the user – it ranges from 55% for users who purchased in the past (promotionally or not) to 85% for users who never made a past purchase. The promotion lasts for three to five days. A user can only make one promotional purchase per cycle, and the length of the promotional cycle is common across all users

Control group

- Users in the control group are not exposed to promotions. They see the in-game currencies within the game shop at the same fixed base price as the treatment group. Everything else is held the same between the treatment and control groups.

Randomization is persistent: i.e., once a user is allocated into a group, she stays in that group for the next 180 days. The pre-fixed bundle configurations, personalized bundle display rule and the exact length of the promotional cycle each active Friday is picked by the firm, and not controlled by us. The firm determined this particular behavioral targeting rule and discounting percentages based on heuristics it developed from past experience (including past experimentation), and was unwilling to allow us to vary these aspects due to the costs associated with such additional randomization. The promotions are, however, representative of promotion schedules offered in app-based games and typical for this industry.

2.2.1 Discussion of empirical approach and treatment effect

With this outline of the experimental setup in place, we briefly discuss the treatment effect measured by the experiment, explain how it relates to the research question and outline the logic for measurement of intertemporal substitution in the context of this specific promotional plan.

What is the treatment effect being measured by the experiment? It is the effect of exposing users to a realistic promotion policy that involves periodic sales for in-game currency. It measures the joint effect of the bundle configuration, bundle display rule, prices, promotional cycle and pop-up based awareness induced by the policy, rather than the separate effect of each of these underlying aspects.

Why is this treatment effect interesting? The reason is that evaluating whether the promotion as a whole is effective is a critical first-step in empirically understanding the impact of intertemporal substitution. If the promotion as a whole were ineffective, a finding of no intertemporal substitution could be criticized as simply arising from its measurement in the context of an ineffective promotional policy. Therefore, we view the measured treatment effect as an important first step to explore the posed research question. Implementing a promotion in any setting, including this one, comprises not just determining the level of price cuts, but also determining the set of products for which price cuts will be implemented, determining the frequency, depth and targetability of the price cuts, as well as developing a strategy to communicate these to relevant consumers. Evaluating whether a promotion is effective requires assessing the combined effect of all these factors on resulting consumer behavior. This is delivered by the measured treatment effect.

Why does it make sense to assess intertemporal substitution effects in the context of such a promotion? The focal problem of empirically understanding the impact of intertemporal substitution in app-based promotions requires comparing consumer behavior in worlds with and without long-term promotions. This problem is not precisely formulated unless one takes a stand on what promotional policy a "no-promotions world" is compared to. One choice for the researcher would be to choose as treatment, a theoretically "optimal" promotional policy that is best from the firm's or social planner's perspective. One complexity of this choice is that such a policy is hard to compute (requiring the solution to a complex intertemporal price discrimination problem). A more serious one is that such policies are unlikely to be practically relevant: Many observed pricing polices are typically found to be very different from their theoretically optimal counterparts (see Cho & Rust, 2010 for an example), risking the criticism that such an experimental comparison does not represent an effect that is likely to be encountered in practice. A different choice for the researcher is to choose as treatment a realistic promotion policy that is representative of what firms in the industry typically implement, and to assess intertemporal substitution effects in the context of this policy. Doing so makes the measured effect and the assessment performed more representative and relevant to the industry analyzed and more likely to be encountered in practice. This is the approach we follow in this paper.

2.2.2 Additional considerations in experimental design

Improving statistical efficiency User attrition for free-to-play games is high, and a large proportion of users tends to drop off after initial trial. The attrition has the potential to induce significant noise into the statistical analysis, reducing precision.[7]

[7] A treated user is exposed to the treatment for 180 days. A user who drops off is exposed to the treatment for $\leq$ 180 days. Thus, the intensity of treatment is lesser for the attrited group. We measure an intent-to-treat effect (ITT), comparing all users who were randomized into treated versus control groups (including those that drop off). The noise in the ITT is higher the higher the number of users who drop off.

To address this, we built the following guardrail into the experimental design. The game has 90 levels. User attrition is highest prior to completing level one. Therefore, we expose users in the treated group to the promotion cycle only after they complete level one in the game. Figure 2 shows this pictorially.

In Fig. 2, a user i in the treated group reaches level one in τ_i days, so is exposed to the treatment for $d_i = 180 - \tau_i$ days. We compare the behavior of all users in the treatment and control groups who have crossed level one. Since users are not exposed to any promotions before they reach level one, the set of users who reach level one in the treatment and control group remain balanced. Figure 11 in the Appendix presents a quantile-quantile plot of the distribution of d_i in the treatment and control groups and shows they are the same, confirming balance across treatment groups.

Sub-treatment conditions Though randomization of the *characteristics* of the promotion sequence (e.g., base price, discount depth) was infeasible, the firm was willing to allow us to vary the timing of exposure of users to promotions. The treatment group described previously is thus created by randomizing users into three sub-groups based on when they start seeing the promotion sequence after download:

1. *Immediate Treatment Group*: This group is exposed to promotions once they complete level one.

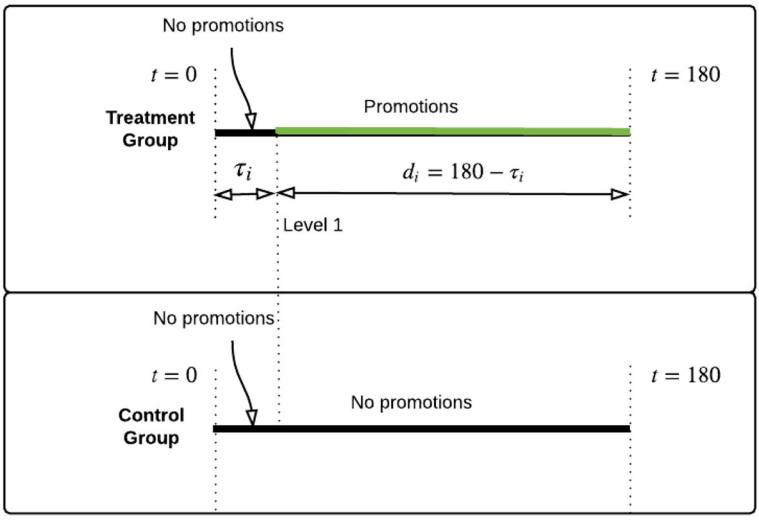


Fig. 2 Experimental Setup. To improve statistical efficiency, users in the treatment group have to complete level one before they are exposed to the promotional schedule, and analysis is based on all users who cross level one. Since users are not exposed to any promotions before they reach level one, the set of users who reach level one in the treatment and control groups remain balanced. A user i in the treated group reaches level one in τ_i days, so is exposed to the treatment for $d_i = 180 - \tau_i$ days. Figure 11 in the Appendix shows that the distribution of d_i in the treatment and control groups is balanced

2. *25-day Delayed Treatment Group*: This group is exposed to promotions once they complete level one *and* it has been at least 25 days since they downloaded the game.
3. *50-day Delayed Treatment Group*: This group is exposed to promotions once they complete level one *and* it has been at least 50 days since they downloaded the game.

Once a user is in a sub-group, he remains in that group for the duration of the experiment.

For most of the initial analysis below, we pool data across the sub-groups to improve statistical efficiency, comparing the treatment group overall, which is exposed to promotions, to the control group which is not. We use comparisons of behavior across the three groups as a way to assess in how far price serves as a signal of quality. The idea is that under price signaling, users who are exposed to price discounts earlier may have a stronger propensity to infer the game is of low quality *ceteris paribus*. Thus, evidence of reduced logins to the app when exposure to promotion is earlier is one indication of signaling effects. We emphasize that a more direct test of price signaling would be to randomize the magnitude of the price discounts which, unfortunately, was not possible.

3 Analysis

A total 160,582 users complete level one across treatment and control groups and remain after data cleaning.[8] The treatment group sizes were decided in collaboration with the firm who had a managerial prior in favor of higher exposure to price promotions. 26,521 users are assigned to the control and 134,061 to the treatment conditions, with 54,883 users allotted to the "immediate" treatment sub-group; 52,652 to the "25-day delayed" treatment sub-group; and 26,526 to the "50-day delayed" treatment sub-group. Tables 7 and 8 in the Appendix report tests of balance across the groups and show that randomization is induced properly.

3.1 Methodological remark

Our analysis approach rests on the single unit treatment value assumption (SUTVA, Rubin, 1978), which we believe to apply here for the following reasons: First, based on an extensive internet search for forum discussions containing "sales,"

[8]The following filters are applied for data cleaning. Data are observed at the device-level. About 50% of devices are connected to one or more `Facebook` accounts. We disregard all devices that are connected to more than one `Facebook` account as this indicates the game was played by more than one user on the respective device. We further disregard devices that are connected to `Facebook` accounts that previously played the game on other platforms, and we disregard devices with `Facebook` accounts that were connected to multiple devices during the experiment. The remaining sample contains devices for which we can reasonably assume that only one user played the game on them. This data was further cleaned for duplicate and obviously broken log entries and for devices that experienced technical difficulties or where the user "hacked" the app during the experimental period.

"promotions," or "offers" conducted after conclusion of the experiment, the three posts included in Fig. 16 cover all posts that touch on "sales." No other posts discussing these matters could be identified. The game further does not contain messaging functionality that would allow players to exchange information with each other. In the authors' opinion, it hence seems unlikely that users' treatment assignment could have been turned instable through user-to-user communication on the bi-weekly sales scheme. The forum discussions further do not contain any mentions by users that feel left out regarding sales or other in-game experiences. While users jointly dissect the game and discuss nuances of their experiences, there is no evidence that users feel treated differently or unfairly. While they hence display a sophisticated consumption style and may play the game strategically, treatment assignment and effects seem to stably materialize at the individual user-level.

Second, in terms of in-game empirical evidence, the login behavior between treated and non-treated groups shown in Fig. 12 remains stable across promotion and non-promotion days. If users in the treatment group without promotions would have been affected by the promotions, one might expect that they either log in more (to see if they are indeed missing out on promotions) or less (as a sign of protest that they are not receiving promotions). In a similar vein, the total number of days that users across treatment groups use the app is stable between groups, see Fig. 13. Figure 8 showing empirical cumulative distribution functions (CDFs) of hours spent in the app by treatment sub-group further supports this notion.

Third, and maybe most convincingly, if SUTVA was violated, one would expect that users in the control group might express their discontent with being excluded from sales by purchasing less when other users receive promotions. As can be seen from Fig. 3 however, purchases and revenue per user in the control group are stable within their confidence intervals for the different days of promotional treatment.

Finally, SUTVA could be violated if a user used several mobile phones to play the game (randomization was performed at the device-level) or played the game on different platforms (i.e., not only their Android phone but also on Facebook on their laptop or desktop), leading them to possibly experience different treatment conditions on different phones. No such cases were reported in customer service. Further, as detailed in footnote eight on page 14, we applied careful data cleaning to remove such cases from the dataset.

3.2 Overall treatment effects

Table 3 reports on the differences in means between treatment and control groups for user free-to-pay conversion (i.e., indicator for purchase of in-game currency over the duration of the experiment); purchases (total number of orders of in-game currency over the duration of the experiment); revenue from premium upgrades (total money spent on in-game currency over the duration of the experiment); advertising revenue and usage (total time spent in app in hours over the duration of the experiment). The main findings are summarized below:

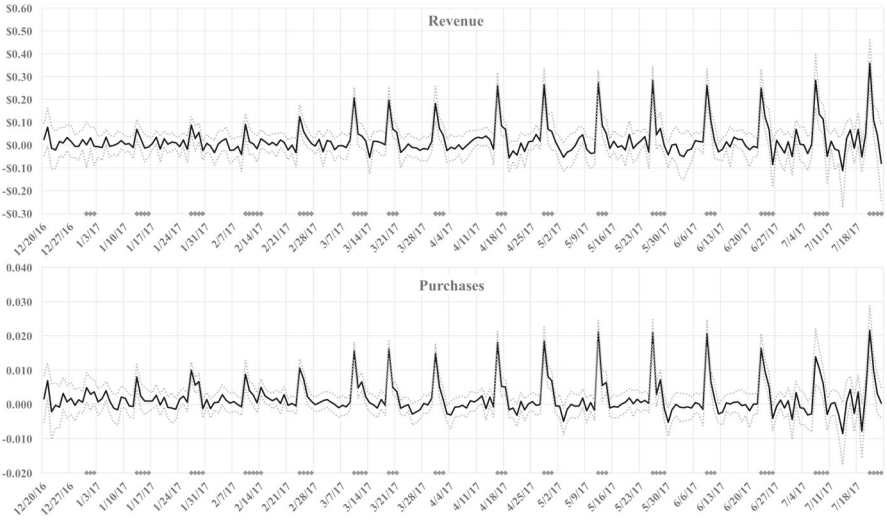


Fig. 3 Mean Difference in Per-day Revenues and Purchases Between the Treated and Control Groups. Mean difference in revenue and purchases per day with 95% confidence band over calendar days between control and promotional treatment groups. Grey diamonds on the date axis indicate days with a promotional cycle

- Free-to-pay conversion increases by 37.1% ($p = 0.000$), from 4.3% in the control group without price promotions to about 6% in the promotional treatment group.
- Purchases of and revenue from premium features increases by 27.2% ($p = 0.000$), and 23.6% ($p = 0.026$), respectively, from an average of 0.29 purchases and \$3.53 in the control group, to 0.37 purchases and \$4.36 in the treatment group.
- Advertising revenue falls by –11.7% ($p = 0.000$) from \$0.29 in the control group to \$0.26 in the treatment group.
- Overall revenue (sum of revenue from premium features and advertising) increases by 20.9% ($p = 0.033$) from \$3.82 in the control group to \$4.62 in the treatment group.
- Game usage (measured as total time spent on the app in hours) falls in the treatment group, but the effect is not statistically significant at the 95%-level.

These results present a favorable picture of promotions as a tool to increase freemium app monetization. Exposure to promotions improves conversion and revenue, with little detectible negative effects on usage and engagement. Even though advertising revenue falls in response to treatment, advertising revenue only makes up about 8% of overall revenue (using values from the control condition); so, overall revenue improves from exposure to treatment due to the increased take-up of in-game purchases. Overall, the promotional strategy that we implemented was effective for the firm.

Table 3 Comparing User Behavior in the Treatment and Control Groups

	Treatment: Promotions N = 134,061		Control: No promotions N = 26,521		Difference (T−C)	*p*-value of Difference
	Mean	SD	Mean	SD		
Free-to-pay conversion (indicator for in-game purchase)	0.059	0.236	0.043	0.204	0.016 ***	0.000
Purchases (no. of orders)	0.37	3.20	0.29	3.09	0.078 ***	0.000
Revenue from premium upgrades (A = a1 + a2)	4.33	57.12	3.53	55.32	0.807 **	0.031
Revenue from non-promoted upgrades (a1)	3.20	45.59	3.53	55.32	-0.329	0.363
Revenue from promoted upgrades (a2)	1.14	15.49	0	0	1.14	-
Advertising revenue (B)	0.255	0.845	0.29	0.921	-0.034 ***	0.000
Overall revenue = A + B	4.59	57.12	3.82	55.32	0.773 **	0.039
Usage (total time spent in app in hours)	179.0	409.5	184.3	419.9	-5.27 *	0.061

The table presents means and standard deviations of intent-to-treat effects on conversion, purchases, revenue and usage. Statistical significance reported using *t*-tests with unequal variance for binary outcomes; *** significant at the 1%-level, ** 5%-level, * 10%-level

The only other paper we know that has implemented a large-scale price/promotion experiment for virtual goods in a video-game is Levitt et al. (2016). Our results about the positive impact of promotions stand in contrast to their results, and we briefly comment on important sources for the differences.[9] In Levitt et al. (2016)'s experiment, pre-existing users of a freemium video-game were randomized into different quantity discount schedules offered on various bundles of in-game currency (gold bars). Though quantity purchased of gold bars was higher under more aggressive discounts, they report no meaningful increases in free-to-pay conversion or revenue in response to discounts. While differences in the nature of the game or in the characteristics of users could explain the difference from our results, one thing to note is that the intensity of price cuts induced in the two experiments is quite different. While the quantity discounts offered in Levitt et al. (2016)'s experiment were large (up to around 70%), they were restricted to bundles of 100 gold bars or more. They report that historically, more than 90% of purchases by the game's users had been for bundles with less than 100 gold bars, so this implies that most of the induced price variation was relevant only for a subset of users who were interested in buying large bundles. In contrast, a feature of the experiment in this paper is the extremely large amount of price variation that is induced across the full range of users. Looking at Table 1, we see that even the smallest in-game currency bundle (of $1.99) is discounted heavily (by 85%), as is the largest bundle (50% off for the $99.99 bundle). This study's experiment induced deep price cuts across a much fuller range of purchasing users than Levitt et al. (2016)'s, which likely explains why the treatment here has a much stronger effect.

3.3 Assessing intertemporal substitution

To assess the extent of intertemporal substitution, Fig. 3 displays a time series over the calendar days of the experiment, of the difference in mean per-day revenues and purchases between the treated and control groups. The calendar days corresponding to a promotion cycle are indicated with grey diamonds on the date axis. The dotted lines represent 95% confidence intervals for the difference in means each day. To see the effects visually, we plot this for active users each day, i.e., those who log in to the app on a given day.[10]

Looking at Fig. 3, we see pronounced regular spikes in both revenue and purchases that coincide with promotional cycles. There is little evidence of reduction prior to a promotion cycle or after one, which would suggest intertemporal substitution. Rather, it seems the spikes reflect expansion in primary demand for in-game currency. Figures 15 and 14 in the Appendix show the same plots separately for the treatment and control groups. Eyeballing these, we can see there is no evidence of systematic demand spikes during the promotional days in the control group, unlike the treatment group.

[9]Thanks to a referee for suggesting this.

[10]Figures 12 and 13 in the Appendix document that there is no difference in the propensity to log in between the treatment and control groups.

To hone in on the days before and after promotions, Fig. 4 pools all the promotion cycles together and displays the mean revenues in treatment and control groups during and around the days of the promotional cycles. The "dashed line" presents the mean in the treatment condition, while the solid line represents the mean in the control condition, and the dotted intervals represent 95% confidence intervals. Looking at Fig. 4, we see there is little difference between the two groups before or after the promotion cycle, suggesting no detectable intertemporal substitution.

Our next assessment is more direct, and leverages the fact that we can observe consumption directly in our data as we can track the consumption of the in-game currencies purchased by users. We first assess whether users spend the in-game currency soon after they purchase it, or whether spending occurs much later, which could be evidence of stockpiling. This analysis is complicated by the way in which in-game currencies are used in the game. Recall, the game has three types of in-game currencies: cash, coins and energy. We observe in the data the redemptions of cash and coins during gameplay as well as the stock of energy available. However, the interpretation of the redemption of coins is quite ambiguous as it is not clear whether the coins redeemed were actually purchased (because coins can also be earned from

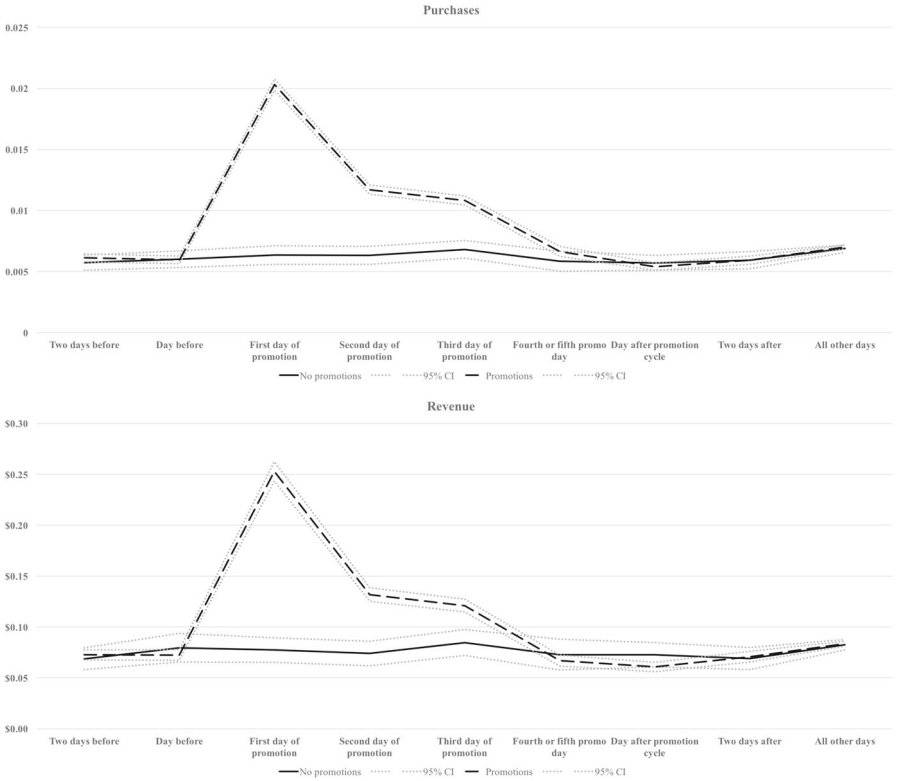


Fig. 4 Mean Per-day Purchases and Revenue in the Treated and Control Groups During and Around Promotional Cycles. Purchases (no. of orders) and revenue (in $) per user per day of and around promotional cycles with 95% confidence band for control and promotional treatment groups

gameplay); or whether it indirectly reflects the redemption of cash (because cash can be used to obtain coins). The interpretation of the redemption of cash is considerably cleaner for this purpose as additional cash is obtained virtually only through real-money in-app purchases. Therefore, we focus on patterns of redemption of cash by users.

Figure 5 shows the difference in means for cash bought and cash redeemed between the two experimental groups. Looking at the top panel in Fig. 5, cash bought is seen to spike during the promotional periods in the treated group relative to the control. The bottom panel of Fig. 5 shows a similar pattern in the cash redeemed, though it is less pronounced.

To hone in on the days before and after promotions, we create an analogous plot to Fig. 4 that pools all the promotion cycles together and displays the mean difference between treatment and control groups in cash bought and redeemed during and around the days of the promotional cycles. This is shown in Fig. 6. Looking at the top panel in Fig. 6, we see cash bought in the treated group statistically significantly rises above that of the control group during the days of the promotion cycles, and is indistinguishable from that of the control group on non-promotion days. Most of the incremental cash purchasing in the treated group relative to the control hence is happening during the promotion cycle. Looking at the bottom panel, we see that cash spent in the treated group rises statistically significantly above that in the control group during the days of the promotional cycle, and the day following it. The cash spent in the treated group two days after the promotion cycle is not statistically significantly different from the cash spent on non-promotion days. This pattern suggests that users spend the incremental cash they buy during the promotional cycle almost immediately within the promotional cycle, and within about a day of the end

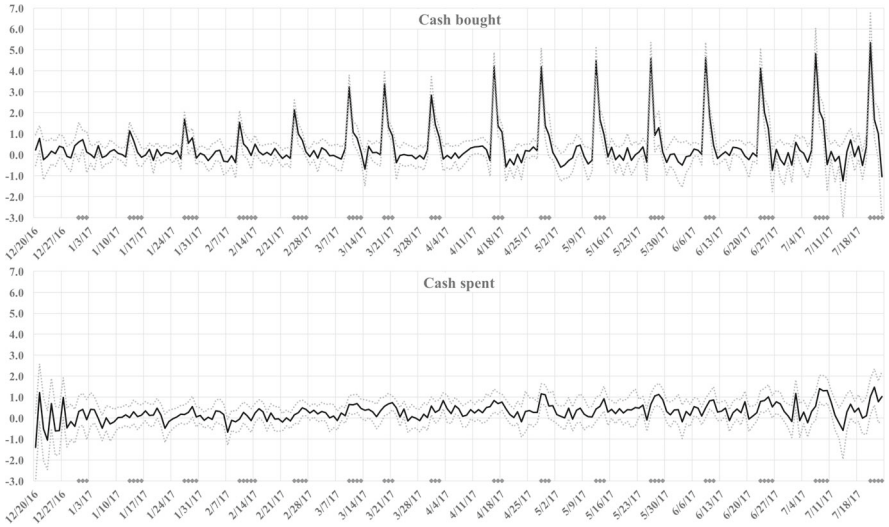


Fig. 5 Difference in Cash Bought and Redeemed Between Treated and Control Groups Over Calendar Days. The figures present the difference in mean cash bought (top) and cash redeemed (bottom) between experimental groups

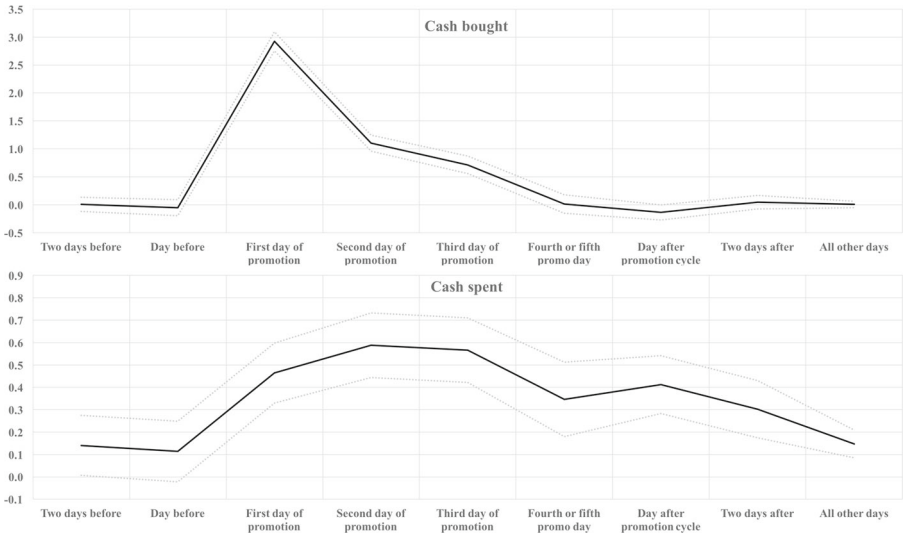


Fig. 6 Difference in Cash Bought and Redeemed Between Treated and Control Groups During and Around Promotional Cycles. The figures present the difference in mean cash bought (top) and cash redeemed (bottom) between experimental groups

of the promotion, with little stockpiling. The immediacy of consumption suggests low regulation of cash spending and little "saving currency for the future."

To more closely assess the presence or absence of intertemporal substitution, we explore in two cuts of the data whether and how the promotional scheme affects the propensity to spend cash out of the cash already purchased. This exercise helps quantify the degree of intertemporal substitution via the rate at which cash is spent out of available cash – in order to avoid intertemporal substitution, this rate would need to remain stable during promotions such that the additional cash purchased in promotions is spent immediately. To implement this analysis, we focus on buying and spending of cash in two samples: [I] During days covered by promotional cycles to compare how promotions affect this rate *between* treatment and control group, and [II] in the treatment group, *within* users. To implement this in regression form, let $\mathcal{P}(t)$ be an indicator of whether day t is part of a promotion cycle, let B_{it} as the cash bought by user i on day t, and compute the total cash bought B_i, and total cash spent S_i by a user i on promotional days in the experiment as,

$$B_i = \sum_{t \in \mathcal{P}(t)} B_{it}$$
$$S_i = \sum_{t \in \mathcal{P}(t)} S_{it}$$

Defining the ratio of daily cash spent during promotional cycles to the daily cash bought during promotional cycles by a user i as $\rho_{i,t} = \frac{S_{i,t}}{B_{i,t}}$, letting i denote an experimental user and t a day of observation, we run a panel regression of the ratio on an intercept, T_i as an indicator for whether user i is in the treatment group, and H_i as an indicator of whether user i is a heavy user (i.e., completed level one on the day of app download, see Table 4 and below in the text for a more detailed definition),

for the subset of all days with promotions, i.e., where $P_t = 1$ (specification [I] in Table 4):[11]

$$\rho_{i,t} = \beta_0 + \beta_1 T_i + \beta_2 H_i + \beta_3 T_i \times H_i \tag{1}$$

Further, we run a panel regression of the ratio on an intercept, H_i as an indicator of whether user i is a heavy user, and P_t as an indicator of promotional days, for the subset of users in the treatment group, i.e., where $T_i = 1$ (specification [II] in Table 4):

$$\rho_{i,t} = \beta_0 + \beta_1 H_i + \beta_2 P_t + \beta_3 H_i \times P_t \tag{2}$$

Looking at results in Table 4, we interpret positive (negative) coefficients as indicating that intertemporal substitution goes down (up) as more (less) cash is spent relative to bought. E.g., considering the first result column, intertemporal substitution on average is lower in the treatment group than in control group (+0.05) during promotional days. Similarly, it is lower on average for heavy compared to non-heavy users (+0.02) and the full model from Eq. 1 in the fifth result column suggests that the lack of intertemporal substitution observed among heavy users during promotions is driven by the promotional treatment (+0.08). Next, turning to the "within"-results, i.e., results for users in the treatment group over time, we can again note a positive coefficient on heavy users (+0.06) but a negative one for promotional days, suggesting that promotions lead to an increase in intertemporal substitution within users (-0.04). However, also in this perspective, heavy users overall are characterized by a positive coefficient (0.08-0.04 = 0.04), suggesting that the overall lack in intertemporal substitution in the purchasing of add-on goods may be driven in particular by heavy users.

3.4 Digging deeper: Why a lack of intertemporal substitution?

What could explain the lack of intertemporal substitution in empirical results? A simple first explanation could be that the quantity of in-game currency included in the promoted bundles is small relative to typical daily consumption. This, combined with the restriction that only one unit of the promoted bundles can be purchased during a promotional cycle, restricts the scope for stockpiling by users of the in-game currencies. To assess this explanation, we look at consumer spending patterns and compare them to the amount of currency available in the promoted bundles. The data suggest that this is unlikely to be the main force driving the results. The mean cash spent per day amongst all users in the control group is 0.977 units. If we consider the subset of control group users who bought cash during the experiment and spent it, the mean cash spent per day is 1.034 units. These are substantially smaller than the cash included in the cheapest promotional bundle (100 units of cash in the

[11] Note that the regression is not exact and ρ_i can sometimes be greater than one because there is no way for us to track reliably whether the cash spent during the promotion is derived fully from the cash bought during promotions. For instance, all users start with an initial endowment of all currencies (including cash), and some of that endowment could be redeemed during a promotional period over and above the cash bought during those periods.

Table 4 Investigating Intertemporal Substitution in Cash Buying and Spending

Dependent Var:	$\frac{\text{Cash Spent}}{\text{Cash Bought}}$											
Specification:	[I]						[II]					
Data includes:	All users, on days with promotions						Users in treatment group, on all days					
	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.
Intercept	1.81	252.28	1.85	604.18	1.82	264.17	1.88	1,095.80	1.90	1,002.38	1.89	951.56
Treatment group	0.05	5.73			0.03	4.31						
Heavy user			0.02	2.01	-0.05	-2.71	0.06	11.99			0.08	12.26
Heavy user × Treatment group					0.08	3.88						
Promotional cycle									-0.04	-10.34	-0.03	-8.36
Promotional cycle × Heavy user											-0.04	-3.42
Nobs	7,754,283								24,265,041			
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	

(User-day) panel data regressions with the ratio of cash spent over cash bought as dependent variable, to measure relative increases (negative coefficient) and decreases (positive coefficient) in the degree of intertemporal substitution. The ratio is regressed on an indicator whether the user is in the treatment group; an indicator whether the user completed level one of the game in zero days ("Heavy user"); an indicator whether the specific day was part of a promotional cycle; and these indicators' select interactions; for the sample of all users on promotional days [I] and for the sample of treated users on all days [II]. T-stat for robust standard errors reported

$1.99 bundle; see Table 1).[12] At these levels, users have ample scope to stockpile the in-game currency included in the promoted bundles for several days of future consumption upon purchase.

Another possible explanation is that the results reflect lack of awareness amongst users about promotions. That is, users are simply unaware that promotions will be offered bi-weekly and hence unable to plan ahead. While plausible, we find this explanation less compelling for our context. The company had been running bi-weekly sales on the game for a long period prior to the experiment. Engaged players discuss details of the game's economy and sales in online forums (see Fig. 16 in the Appendix). Furthermore, the use of in-game promotions is a regular feature of free-to-play games in this category and most users, especially those who tend to purchase in-game currencies, are likely familiar with the category. It is therefore likely they are aware of the regular bi-weekly occurrence of sales.

3.4.1 Positive covariation in valuations, complementarities plus freemium

Based on our results, we explore alternative mechanisms to explain the observed lack of intertemporal substitution. Our preferred explanation is simple, relying on three aspects that characterize the typical demand profile and the monetization scheme of many apps: First, the valuation of users for consumption of base and premium app features tend to positively covary in the target population; second, the base and premium features tend to be complements; and third, consumption of the base good is free. By the first statement, we mean that users who tend to value consuming the base good more, also tend to value consuming the add-on premium features more (as, e.g., observed in Appel et al., 2020). This assertion implies that user valuations across goods are positively correlated which is the default presumption in gaming: For example, "heavy users" or "hardcore gamers" who value highly playing the base game, also tend to highly value the add-on premium features of the game that they can obtain by spending in-game currency.[13] By the second aspect, we mean that marginal consumption of the base good makes additional consumption of the add-on good more attractive; and/or marginal consumption of the add-on good makes additional consumption of the base good more attractive. This complementarity is the natural presumption in gaming and other settings using a freemium pricing scheme: additional gameplay (consumption of base good) often makes premium features (the add-on good) more desirable, and utilizing premium features enhances the value of playing the game or using the base good. Indeed, game environments and freemium apps, e.g., in dating (Tinder, Hinge), lifestyle (Runtastic, Calm) or productivity (Dropbox, Zoom), are often explicitly designed by firms to induce such complementarity. By the third aspect, we refer to the fact that a freemium-based

[12] We report these for cash rather than energy or coins because its interpretation is cleaner, as discussed above and also later in this section.

[13] The reverse could happen in a perverse situation where those that value highly playing the base game value its premium features less. Then, valuations would be negatively, rather than positively, covarying across users.

monetization model implies a base good price of zero and is available to users for free. In our context, the base game is always available to users to play for free.

In such a situation, consider a price reduction in the in-game currency such as the one induced by the experimental promotion, which causes some users to buy more add-on goods on the margin. Because of the positive covariation in valuations, those who tend to buy more of the add-on goods, will also tend to be those that value the base good more. These users will tend to consume the base good more. Alternatively, if complementarities exist in utility, those who are induced to buy more of the add-on good on the margin will value consuming the base good more, inducing them to consume it more. It is possible that in any single context, both of these are at play simultaneously. Conventionally, the fact that the base good has a non-zero price would be an impediment to the additional consumption induced for the base good. However, under the freemium model of monetization, the base good is available for free − so its consumption is costless, which leads to higher levels of incremental consumption of the base good than a conventional situation where the base good also has positive prices. As the base good is consumed by these users, the opportunities and interest in acquiring more add-on goods expands, driven by the fact that consumption of premium features occurs in the context of the consumption of the base good. This is a form of complementarity whereby the incremental consumption of the base good enhances the value of add-on goods to the user. This leads to additional utilization of add-on goods, and so on, forming a feedback cycle, culminating in high levels of both gameplay and in-game currency utilization in response to the promotion –making it difficult for users to regulate their consumption of in-game currencies in promoted periods sufficiently (and stockpile for future high-price periods) to generate the kind of intertemporal substitution required for sales to be unprofitable.

3.4.2 Exploratory evidence

We explore whether there is any support for this simple explanation in the data. Our empirical strategy is as follows: We isolate a set of users who a priori are expected to have high preference for the base good. Then, we ask whether those users tend to consume more of both the base good and the add-on goods in response to the promotion. The basic idea is that if valuations are positively correlated as postulated above, users who buy more of the in-game currency (and thus of add-on goods) in response to the promotion, will tend to be the ones who value consumption of the base game more. These users should also show higher gameplay during the promotion. Therefore, both their consumption of the base good and the add-on good should increase sharply during promotional periods.

If complementarities exist, it will accentuate this effect. The complementarities induce feedback between add-on good consumption and gameplay. If such feedback occurs, it would be higher for those that buy more add-on goods or have higher gameplay in response to the promotion. With positively correlated valuations, those that buy more add-on goods will also have higher gameplay; and in turn will also have the highest feedback effect. So, the existence of complementarities will make their consumption of the base good and the add-on good increase even more during promotional periods.

How should we think of this empirical test? This assessment explores the joint existence of positively covarying valuations and complementarities, rather than isolating the effect of either separately. Our goal here is to offer a simple mechanism that can rationalize our results along with plausible empirical support consistent with that mechanism; separately isolating one piece of the phenomena (heterogeneity) from the other (complementarity) is not required for this goal. The experiment was not designed explicitly to sort between theories; and hence, more deeply testing formal theories of consumption in freemium and exploring the source of complementarities and/or covariance in valuations, or isolating them separately, is beyond the scope of this paper.

Patterns in purchase and consumption of add-on goods We start with exploring purchases of in-game currency, testing whether "heavy users" − who value the game more − are the ones who buy more in-game currency during promotional cycles. Here and in the analysis below, we proxy for "heavy users" using the time taken to reach level one in the game. Heavier users are expected to reach level one faster. Since randomization into experimental groups is done only after users reach level one (as explained in more detail below), this forms a valid pre-experimental baseline characteristic on the basis of which to split subsequent post-treatment purchase behavior.

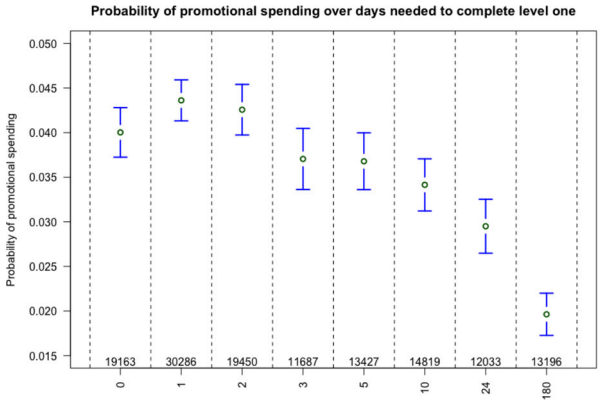
Figure 7a presents plots of the probability of spending money on a promotion during the experiment as a function of the time taken to reach level one. This plot is generated for users in the treatment group (because only treated users are exposed to the promotion cycles). Figure 7b shows the same plot for the amount of money spent on promotions for those who purchased something on promotion during the experiment (i.e., spend | spend $>$ 0), again for users in the treatment group. Both are found to be higher for those who reached level one faster. This suggests "heavy users" are the ones who buy more in-game currency during promotional cycles.

The next set of analyses focuses on consumption of in-game goods, rather than purchases. Based on the discussion in the previous section, we focus on consumption of the most exclusive in-game good, "cash." In the previous section, we documented that cash spent spikes in the treated group during promotion cycles. We test whether the increase in cash spent observed in the treated group during promotion cycles is driven by heavy users. Letting i denote an experimental user, t denote day, T_i be an indicator for whether user i is in the treatment group, H_i be an indicator of whether user i is a heavy user, P_t be an indicator of whether day t is a promotional day, and S_{it} denote the cash spent by user i on day t, we run the following panel-data regression,

$$S_{it} = \beta_0+\beta_1 T_i+\beta_2 H_i+\beta_3 P_t+\beta_4 T_i\times H_i+\beta_5 T_i\times P_t+\beta_6 H_i\times P_t+\beta_7 T_i\times H_i\times P_t \quad (3)$$

Column [C] in Table 5 presents the results for the full model, and columns [A] and [B] present results for relevant subsets of the independent variables. Results show that heavy users tend to spend more cash in general, but especially so when treated: The treatment effect for a heavy user (0.10 units of cash spent per day for the full model) is about five times the treatment effect for a non-heavy user (0.02). Column [D] in Table 5 runs the same regression but restricting the days in the regression to only those during the promotion cycle (and hence dropping variable P_i), so the coefficients in the regression represent the increase in cash spent induced by treatment during

(a) Extensive Margin



(b) Intensive Margin

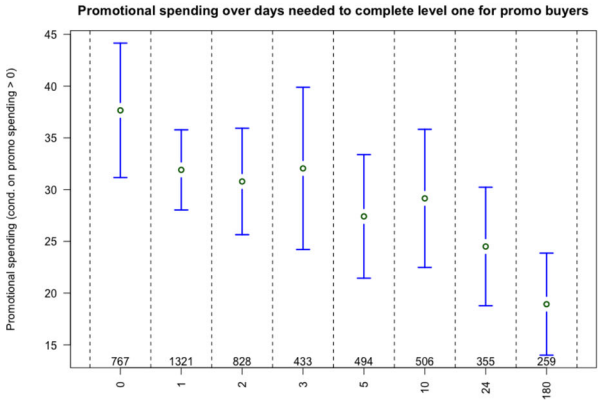


Fig. 7 Promotional Spending as a Function of Time Needed to Reach Level One. The figures present the probability of spending money on promotions during the experiment (top), and the amount of money spent on promotions conditional on non-zero promotional spending during the experiment (bottom) as a function of the time (in days) needed to reach level one

promotion cycles. Looking at the results, we see the treatment effect for a heavy user (0.18 unit of cash spent per day) is about twice the treatment effect for a non-heavy user (0.09) during the promotion cycle. Both these show that heavy users respond more in terms of consumption of the premium in-game good compared to non-heavy users: i.e., when promotions occur, the users who tend to value the base good more are the ones who consume more of the add-on good.

Patterns in the consumption of the base good We now analyze patterns in consumption of the base good in response to promotions. We operationalize consumption of the base good in the data by gameplay, defined as the number of seconds played by a user each day of the game.

First, to establish a baseline, we test whether time spent playing the game spikes during the days of the promotional cycle in the treated group. We analyze this using a

Table 5 Investigating Cash Spent by Heavy vs. Non-Heavy Users

Dependent Var:	Cash Spent							
Specification:	[A]		[B]		[C]		[D]	
Data includes:	All periods						Periods of promotional cycle	
	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.
Intercept	0.96	203.76	0.99	189.40	0.98	176.33	0.93	102.15
Treatment group	0.04	7.52	0.03	5.83	0.02	3.21	0.09	9.19
Heavy user	0.09	6.13			0.11	6.44	0.03	1.08
Heavy user × Treatment group	0.12	7.53			0.10	5.34	0.18	5.79
Promotional cycle			-0.05	-5.42	-0.04	-4.04		
Promotional cycle × Treatment group			0.08	7.55	0.07	6.24		
Promotional cycle × Heavy user					-0.08	-2.54		
Promo c. × Heavy u. × Treatm. group					0.08	2.13		
Nobs	29,065,342						7,754,283	
R^2	0.00		0.00		0.00		0.00	

Models [A], [B] and [C] are (user-day) panel data regressions of cash spent by a user on a day on an indicator of whether the user reached level one of the game in zero days ("Heavy user"); an indicator of whether the user is in the treatment group; an indicator if the specific day was part of a promotional cycle; and interactions of all indicators. [A], [B] and [C] run the regression on all user-day observations in the experiment; [D] runs the same regression on only the days of the promotional cycle. T-stat for robust standard errors reported

panel-data regression analogous to Eq. 3, regressing the gameplay by a user i on day t on an indicator for whether user i is in the treatment group, an indicator for whether day t belongs to a promotional cycle, and interactions of the two. Column [B] in Table 6 shows the results. Gameplay is seen to spike by about 543 seconds per day on average (about nine minutes) in the treated group during days of the promotional cycle.

Next, we focus on heavy users, and ask whether their time spent playing the game spikes similarly when treated (because they are exposed to promotions); and if it does, whether the spike in time spent playing the game when treated is higher during the days of the promotional cycle (when they buy more add-on goods). To do this, we run an analogous regression to the one above, regressing the gameplay by a user i on day t on an indicator for whether user i is in the treatment group, an indicator for user i is a heavy user, and interactions of the two. Column [A] in Table 6 presents the results, and shows that heavy users tend to play about 861 seconds more per day on average in gameplay (about 14 minutes) when treated. Column [C] finally runs the regression on the full set of independent variables, corroborating the previously reported effects and showing that heavy users spend another 183 seconds a day playing during promo cycle days when exposed to the promotional treatment.

Column [D] runs the same regression but restricting the days in the regression to only those during the promotion cycle. We see that heavy users tend to play even more during days of the promotion cycle when treated: about 996 seconds more per day on average in gameplay (about 17 minutes).

A take-away from these results is that, in the vicinity of a price shock to the add-on good — i.e., the promotion, consumption of the base game also spikes, especially for heavy users who a priori are expected to like the base good more. Overall, we take this behavioral response as consistent with a demand profile characterized by positively covarying valuations and/or complementarities.

3.5 Assessing price as a quality signal

A final analysis involves assessing whether promotions serve as negative signals of quality about the base game. For this analysis, we compare game usage, measured as time spent in the app. If users perceive the game has lower quality in response to seeing it frequently discounted, we expect they may reduce usage or stop playing the game. This would be reflected in the observed time spent by users in the treatment and control groups. Tables 3 and 6 already noted that there is no statistically significant difference in mean time spent between treated and control groups. Figure 8 plots the empirical CDF of the time spent in the app for the three sub-treatments of the treated group, in which the onset time of promotions was varied (as described in Section 2.2.2). There is no discernible difference in time spent across the three sub-groups. If users who are exposed to promotions sooner and more would take this as a signal of low quality, we would expect their propensity to log into the app and to spend time in the app to go down – neither is the case as shown in Fig. 8 for time spent and in Figs. 12 and 13 in the Appendix for login behavior.

For added confirmation, Fig. 9 looks at conversion, purchases, revenue, and ad revenue across the three sub-groups. Generally, the patterns go in the opposite direction

Table 6 Investigating Time Spent by Heavy vs. Non-Heavy Users

Dependent Var:	Time Spent on Gameplay (in sec.)							
Specification:	[A]		[B]		[C]		[D]	
Data includes:	All periods						Periods of promotional cycle	
	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.	Coeff.	$t-$stat.
Intercept	4,991.91	24.93	4,833.54	21.03	5,116.06	19.06	4,650.92	33.61
Treatment group	-316.31	-1.46	-336.96	-1.36	-454.69	-1.58	63.79	0.34
Heavy user	-1,789.98	-8.88			-1,964.96	-7.29	-1,308.82	-8.99
Heavy user × Treatment group	861.26	3.32			812.90	2.54	995.68	2.39
Promotional cycle			-370.22	-1.43	-465.14	-1.54		
Promotional cycle × Treatment group			543.68	1.81	518.48	1.51		
Promotional cycle × Heavy user					656.14	2.14		
Promo c. × Heavy u. × Treatm. group					182.77	0.35		
Nobs	29,065,342						7,754,283	
R^2	0.00		0.00		0.00		0.00	

Models [A], [B] and [C] are (user-day) panel data regressions of gameplay by a user on a day (in seconds) on an indicator of whether the user reached level one of the game in zero days ("Heavy user"); an indicator of whether the user is in the treatment group; an indicator if the specific day was part of a promotional cycle; and interactions of all indicators. [A], [B] and [C] run the regression on all user-day observations in the experiment; [D] runs the same regression on only the days of the promotional cycle. T-stat for robust standard errors reported

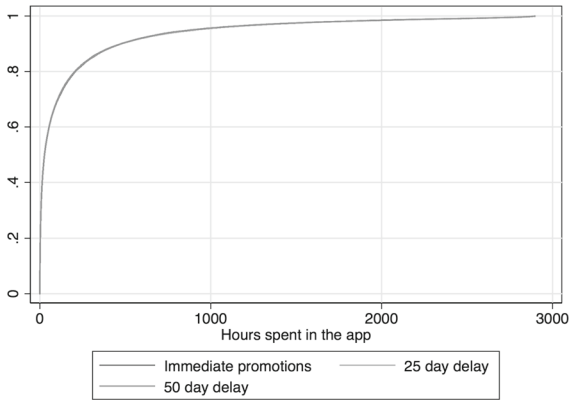


Fig. 8 Empirical CDFs of Hours Spent in the App by Treatment Sub-group. The plot shows the empirical CDF of total hours spent in the app for the three promotional sub-groups. All three CDFs are plotted, but are visually indistinguishable from each other

of an adverse quality signaling story: Conversion and purchases over the long-run are higher the earlier promotions begin.[14] Overall therefore, these results do not provide compelling evidence for meaningful harmful consequences of sales to induce negative inferences about game quality.

4 Concluding discussion

Overall, our experimental results paint an attractive picture for the use of sales in freemium-based app monetization: Conversion to premium, purchases, and revenue increase with little reduction in usage. While advertising revenue is reduced, in this setting, advertising forms a small component of overall revenue, so its economic significance is limited. However, if advertising takes a higher stake in overall monetization, this effect may dominate economically and, therefore, we think this finding is of separate interest.

We postulated that the zero price of the base product that makes its consumption virtually costless, combined with the complementarity between the base product and premium features and/or positive correlations in valuations for the goods across users, can help explain these results. In gaming in particular, complementarities may be stronger than other app settings due to the "addictive" nature of the gameplay, and this effect may be especially salient for the subset of hardcore gamers that pay substantially for games. While it is not unreasonable that most freemium base and premium goods are complements, the quantitative significance of such complementarities in other settings is an empirical question and warrants further, careful empirical testing

[14] We also see in Fig. 9 that advertising revenue is lower when price promotions begin earlier. Given the patterns in Fig. 8, this is not driven by a mechanical linkage of advertising exposure and advertising revenue to time spent in the app. The fact that early onset of promotions reduces advertising revenue without changing time spent in the app significantly, suggests that promotions and advertising are substitutes from the perspective of users. This can occur for instance, because both provide ways to obtain in-game currencies, or because both compete for scarce user attention. There is some evidence in the literature that promotions can serve as advertising (e.g., Sahni et al., 2016 and the literature cited there), so it is possible that promotions tend to deplete a "mental account" associated with attention to advertising.

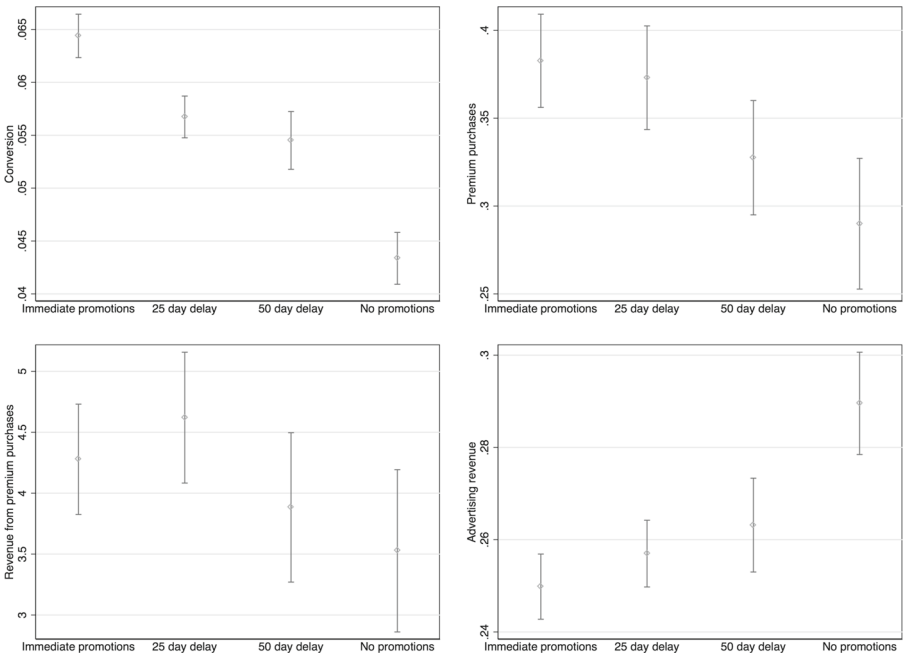


Fig. 9 Impact of Different Promotion Onset Times on Monetization Outcomes. The charts show mean with 95% confidence interval for key monetization outcomes across promotional sub-treatments that vary the onset time of price promotions after users' adoption of the freemium product, and for the control condition

before wider conclusions can be drawn or generalized. Also valuable would be experiments that vary the characteristics of the sales strategy (depth, frequency, items), which would enable studying the empirical consequences of these more deeply and help in the formulation of better policies. The role of addiction and self-control for such goods and their implications for pricing strategy would also be interesting areas of future study to extend current literature considering usage restrictions (Nevskaya and Albuquerque, 2019; Jo et al., 2020).

Speaking to the effect of the immediacy of the utility gain from the base and premium complementarity, the results of this study may have normative implications for the design of freemium experiences similar to suggestions by managerial work (Eyal, 2014; Alter, 2017): Purposeful product design can create and reinforce the conjectured complementarity by ensuring that users experience a strong increase in utility immediately after they make a premium purchase. This can be achieved, e.g., by progressing users substantially through a level (games), matching them with much better mates (dating) or professional connections (networking), or substantially increasing the curation and recommendation quality of content (music, movies, news) immediately after a purchase. The stronger the complementarity between free and premium experience and the more immediate the gain in utility from adoption of premium content, the more effectively can regular promotional activity be expected to incrementally increase overall demand.

Appendix

In-game currency and sales in `Candy Crush Saga`

(a) In-Game Currency (b) Sale of In-Game Currency

Fig. 10 Examples of In-Game Currency and Sales in the Popular Video Game `Candy Crush Saga`. The left panel shows a screen shot of "gold bars", a type of in-game currency available for purchase in the game `Candy Crush Saga`. The user can buy the bars in various bundles. The right panel shows a sale run in the game for a bundle of in game currencies and other in-game features

Days in app is balanced between treatment and control groups

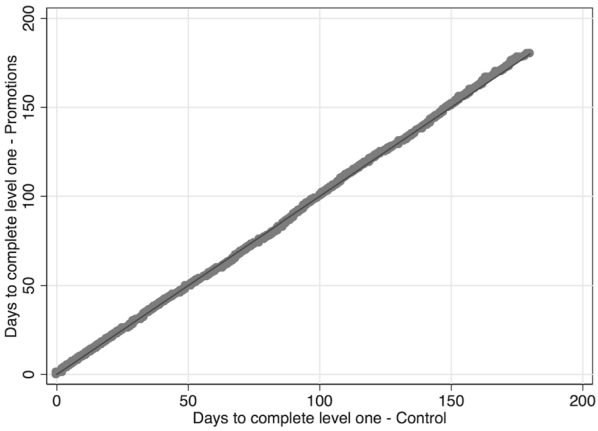


Fig. 11 Q-Q Plot of Days to Complete Level One in Treated and Control Groups. Users in the treatment group have to complete level one before they are exposed to the promotional schedule. Since users are not exposed to any promotions before they reach level one, the set of users who reach level one in the treatment and control group remain balanced. Suppose a user i in the treated group reaches level one in τ_i days, so is exposed to the treatment for $d_i = 180 - \tau_i$ days. The figure presents a quantile-quantile plot of the distribution of d_i in the treatment (y-axis) and control (x-axis) groups to check balance. The distribution of d_i in the treatment and control groups is seen to be balanced

Balance tests: Treatment vs. control groups

Table 7 Pre-treatment Tests of Balance for Treatment and Control Groups

	Treatment: Promotions (T)		Control: No promotions (C)		
	N = 134,061		*N* = 26,521		
	Mean	SD	Mean	SD	*p*-value
Days to complete level one	9.88	(22.42)	9.84	(22.43)	0.771
Hours spent in app	8.47	(20.82)	8.52	(21.16)	0.756
Conversion	0.006	(0.078)	0.006	(0.074)	0.256
Purchases	0.008	(0.128)	0.007	(0.121)	0.192
Revenue	0.072	(1.38)	0.064	(1.35)	0.391
Connected to Facebook	0.295	(0.456)	0.300	(0.458)	0.131
Device memory	1,860.3	(869.5)	1,862.4	(869.3)	0.718
Device DPI (dots-per-inch)	303.6	(120.6)	303.3	(120.2)	0.701
Downloaded game in the US	0.32	(0.466)	0.32	(0.466)	0.92
Downloaded game in the UK	0.050	(0.218)	0.049	(0.217)	0.751
Downloaded game in Germany	0.153	(0.360)	0.154	(0.361)	0.728
Downloaded game in France	0.106	(0.307)	0.106	(0.308)	0.738

Pre-treatment indicators that are observed during users' play of level one of the game for control and treatment groups with *p*-values of *t*-test for difference

Balance tests: Three treatment sub-groups

Table 8 Pre-treatment Tests of Balance for Treatment Sub-groups

	Treatment: Immediate Promotions (T0)		Treatment: 25 Day Delay (T25)		Treatment: 50 Day Delay (T50)		
	N = 54,883		*N* = 52,652		*N* = 26,526		
	Mean	SD	Mean	SD	Mean	SD	*p*-value
Days to complete level one	9.83	(22.32)	9.90	(22.33)	9.95	(22.80)	0.773
Hours spent in app	8.55	(21.17)	8.49	(20.31)	8.26	(21.08)	0.172
Conversion	0.007	(0.081)	0.006	(0.077)	0.006	(0.074)	0.167
Purchases	0.009	(0.144)	0.008	(0.120)	0.007	(0.110)	0.034
Revenue	0.078	(1.51)	0.070	(1.33)	0.061	(1.16)	0.233
Connected to Facebook	0.296	(0.456)	0.295	(0.456)	0.295	(0.456)	0.973
Device memory	1,855.1	(871.0)	1,862.6	(866.5)	1,866.5	(872.6)	0.159
Device DPI (dots-per-inch)	302.5	(120.4)	304.4	(120.7)	304.2	(120.8)	0.025
Download in US	0.320	(0.466)	0.321	(0.467)	0.320	(0.467)	0.900
Download in UK	0.049	(0.215)	0.052	(0.221)	0.050	(0.217)	0.063
Download in Germany	0.156	(0.362)	0.151	(0.358)	0.151	(0.358)	0.090
Download in France	0.104	(0.306)	0.107	(0.309)	0.106	(0.308)	0.396

Pre-treatment indicators that are observed during users' play of level one of the game for all treatment sub-groups. *p*-values are from three-way ANOVA tests for differences

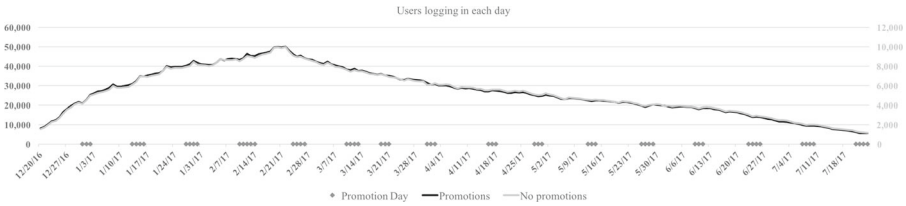


Fig. 12 Number of Users Logging into the App by Day Split by Group

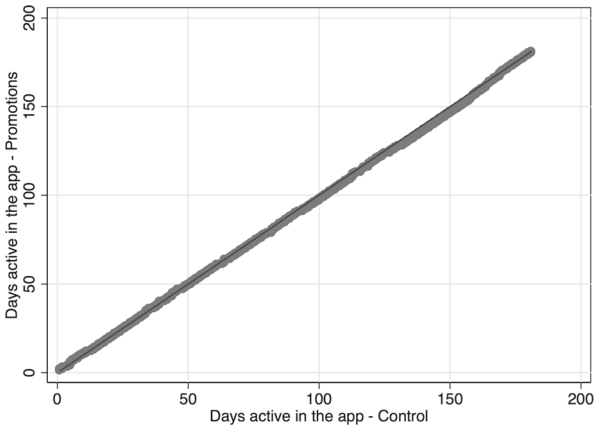


Fig. 13 Q-Q Plot of Number of Days a User is Active in the App Split by Group. Days active is defined as the number of days (out of 180) a user logged into the game. The distribution of login propensity is seen to be the same across the treatment and control groups

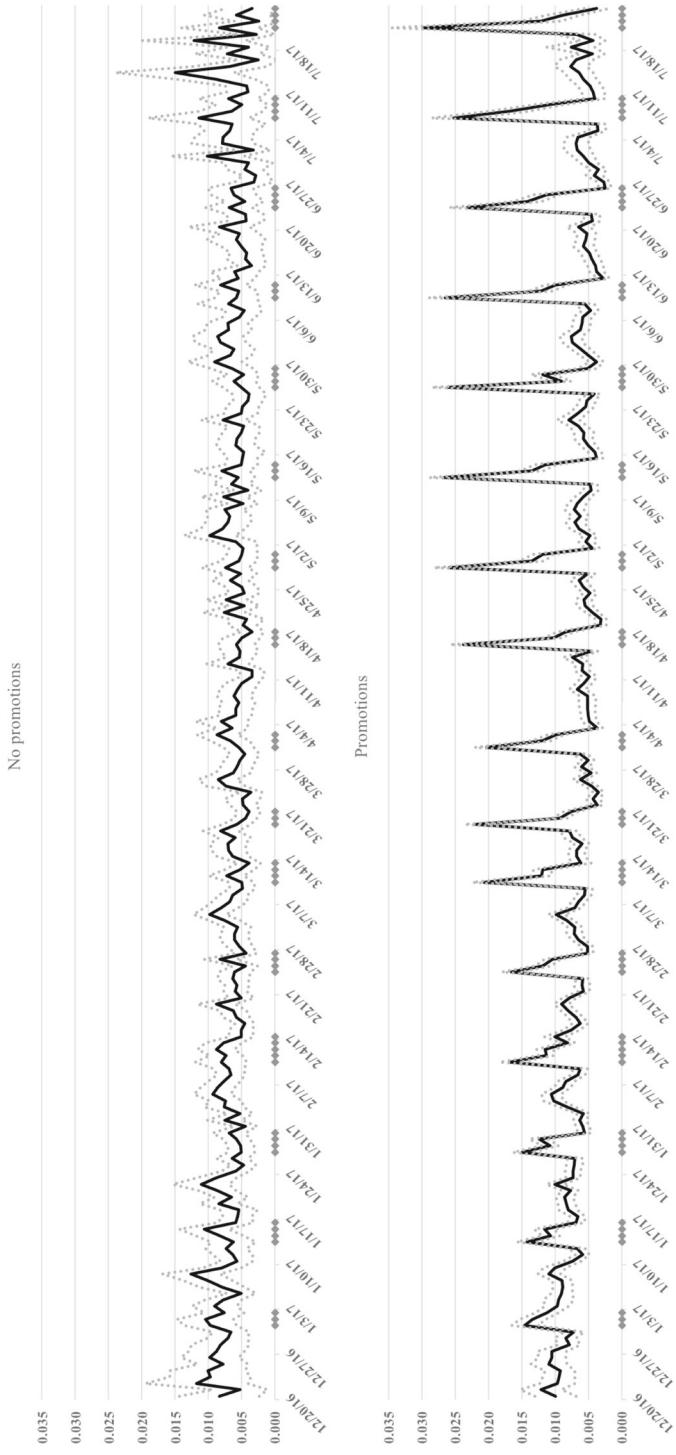


Fig. 14 Per-day Purchases for Treated and Control Groups. Purchases per day with 95% confidence band over calendar days. Grey diamonds on the date axis indicate days with a promotional cycle

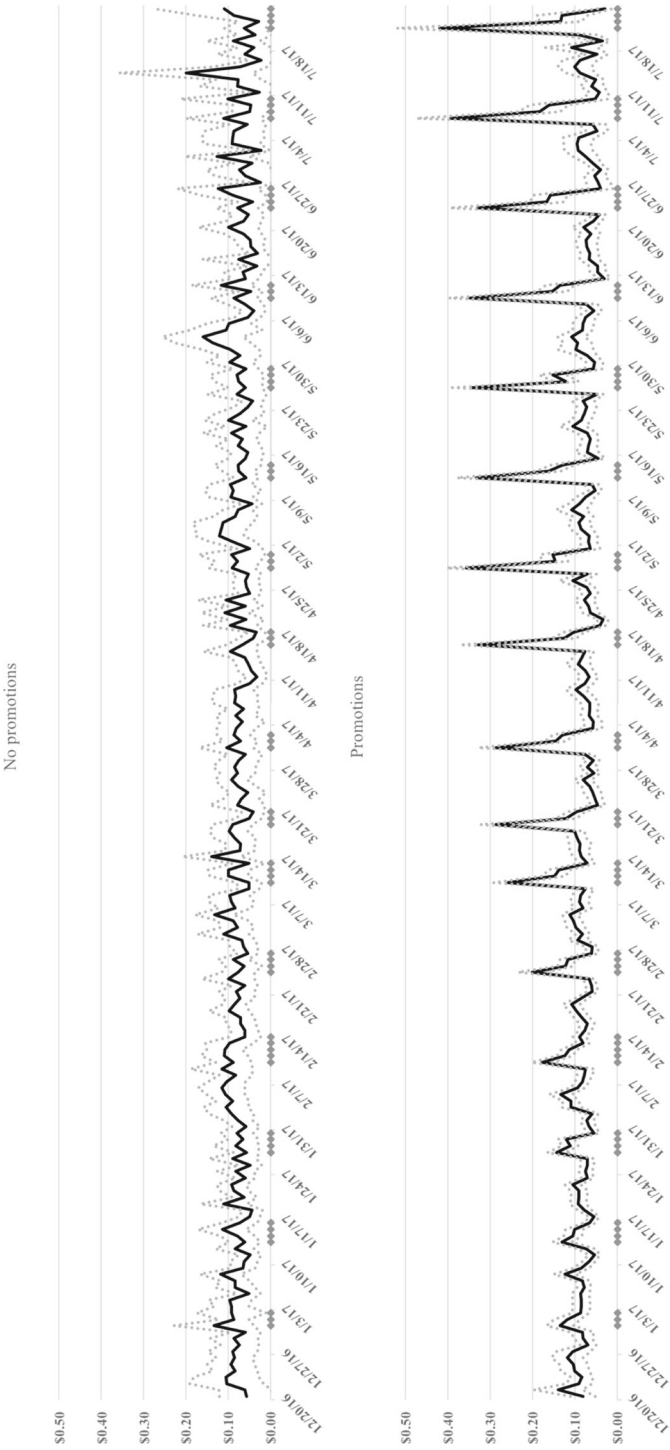


Fig. 15 Per-day Revenues for Treated and Control Groups. Revenue per day with 95% confidence band over calendar days. Grey diamonds on the date axis indicate days with a promotional cycle

Information about sales is widely discussed on online gaming forums

Fig. 16 Screen Shots from a Gaming Forum where Information about Sales on the Video Game Is Discussed Prior to the Experiment

Declarations

Conflict of Interests The first author worked in part-time employment as a research scientist at the industry collaborator while data for the study were collected.

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Affiliations

Julian Runge[1,2] · Jonathan Levav[3] · Harikesh S. Nair[3]

Jonathan Levav
jlevav@stanford.edu

Harikesh S. Nair
harikesh.nair@stanford.edu

1 Fuqua School of Business, Duke University, 100 Fuqua Drive, Durham, NC 27708, USA

2 Institute for Marketing, RWTH Aachen University, Kackertstr. 7, 52072 Aachen, Germany

3 Stanford Graduate School of Business, Stanford University, 655 Knight Way, Stanford, CA 94305, USA

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